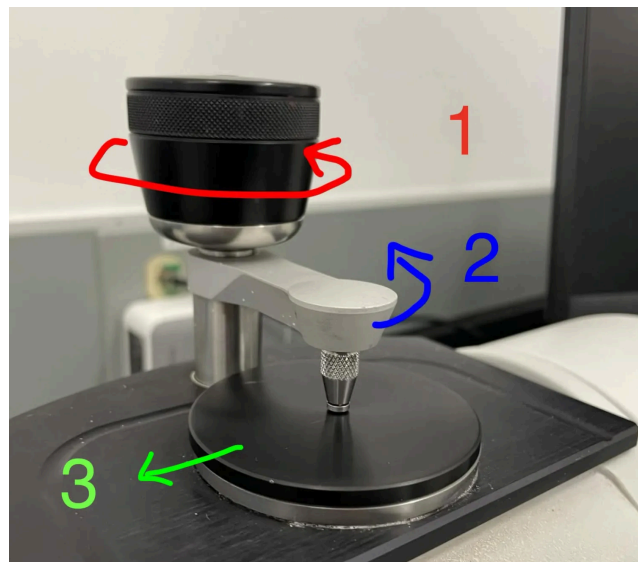


3.000 - FTIR Station

In this rotation you will compare the water content of coffee beans at various roast levels using the FTIR. *You will need to save the relative height of the water peak of each bean as a percent, calculated as explained later. You will submit this to Canvas once this rotation is complete.*

Instructions:

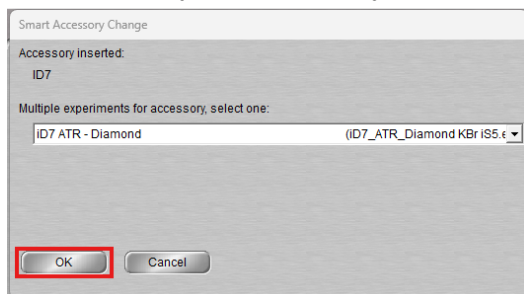
1. Rotate the black knob counterclockwise a few turns. Then, rotate the gray arm a quarter turn to the right (counterclockwise). Finally, remove the black cover.
2. Clean the sample stage with a Kimwipe and isopropyl alcohol. In the center of the stage, there is a small diamond through which the infrared light comes from. Pay special attention to making sure that's clean.



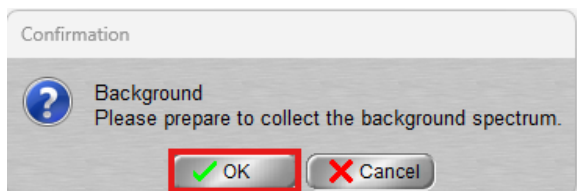
3. Log into the computer using your kerberos.

4. Open OMNIC ().

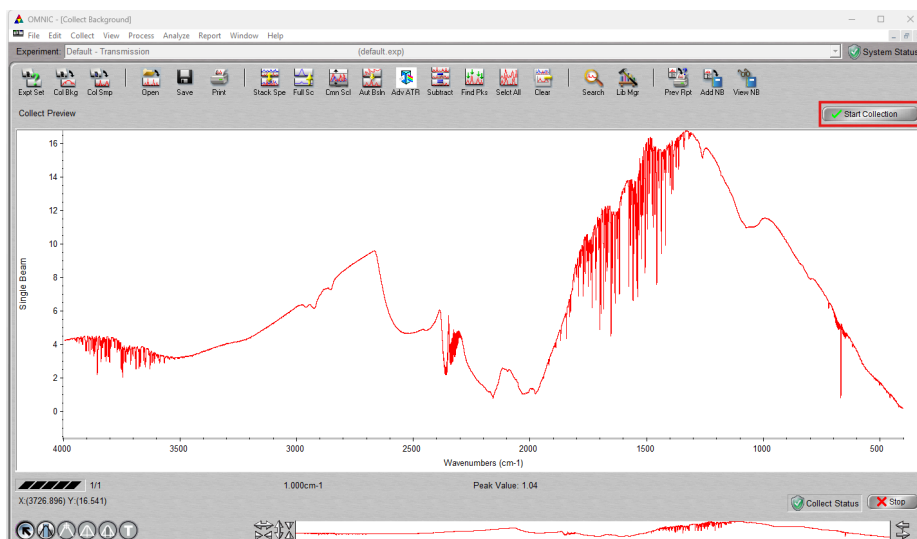
5. When it asks you to confirm your accessory, click OK.



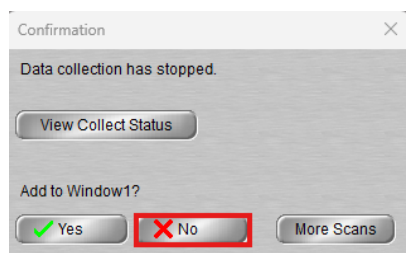
6. Collect your background measurement, by clicking Col Bkg().
7. Click OK to confirm that you're ready to collect the background.



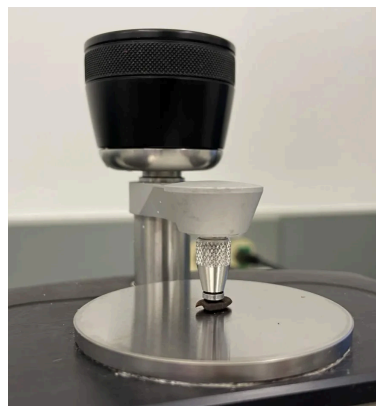
8. Now, you're in a real time display of the background measurement. This is a chance to reclean your sample stage if the background looks bad. Below is what the background should look like. Click "Start Collection" once happy.



9. Wait for the spectrum to finish. In the meantime, read the FTIR info sheet attached to the end of this packet.
10. When it prompts you to add the background to the window, click No.



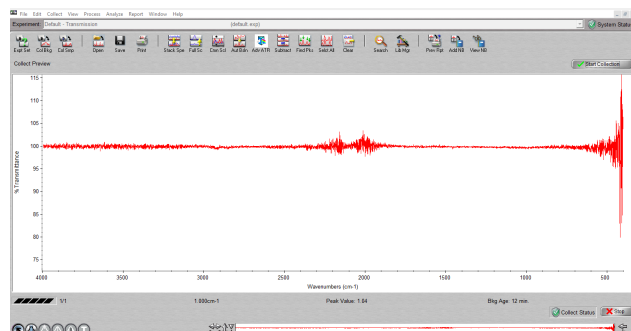
11. Now you're ready to record the FTIR spectrum of your beans. Put a bean on the sample stage. Your goal is to cover the diamond as best as possible. It is your design choice on how you want to do that (whole bean, cut a piece off the bean, grind the bean, etc.)
12. Once you put the bean on the sample stage. Rotate the gray arm back over the bean, and rotate the knob clockwise to hold your sample down. Do not worry about adding too much pressure, as the knob will release once it gets to a certain pressure.



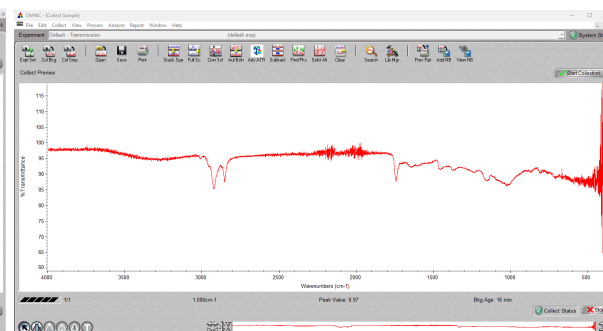
13. In OMNIC, click "Col Smp" () and name your sample.

14. In the real time display page, you have the chance to improve your spectrum by repositioning/grinding/cutting your bean. Below are examples of good and bad spectrums.

Bad spectrum:



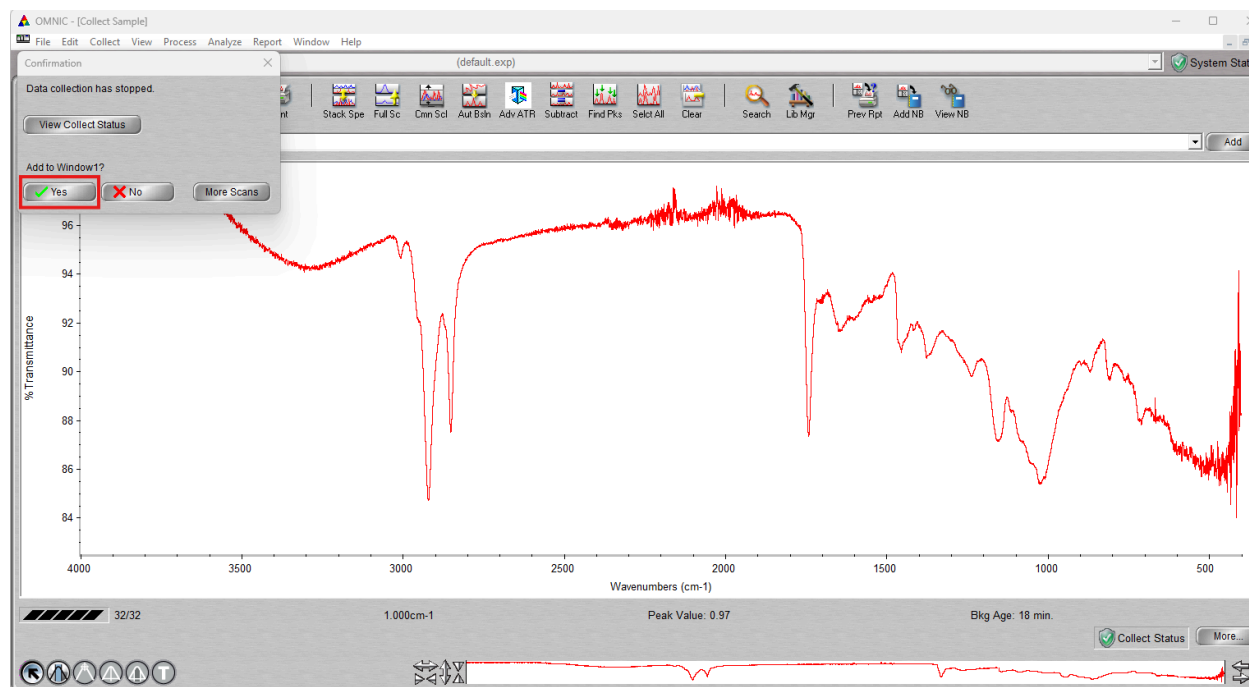
Good spectrum:



15. Once you've achieved a good enough spectrum, click Start Collection.

16. Wait for the measurement to finish. In the meantime, use the "Infrared Spectroscopy Absorption Table" to guess which peaks correspond to which bonds.

17. Once your sample is complete, click OK when prompted to add it to the window.



18. Save your spectrum as a Spectra (.SPA) file by going to File > Save As.

19. Remove your sample and clean the stage.

20. Repeat steps 12-20 for the beans of differing roast levels in front of you.



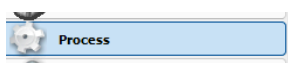
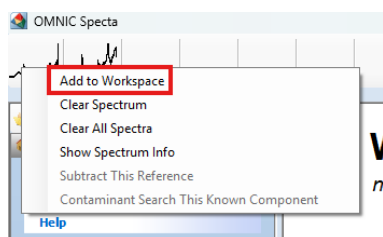
21. Now it's time for data analysis. Open Omnic Spectra ().

- a. Omnic Spectra is a powerful software that allows you to process your spectra and compare it to a database of spectra to identify mystery samples. If you have time left after the activity, a TA can show you how to access these features.



22. Click the little file symbol () and add your spectra.

23. Right click your spectrum in the upper left corner and click Add to Workspace.

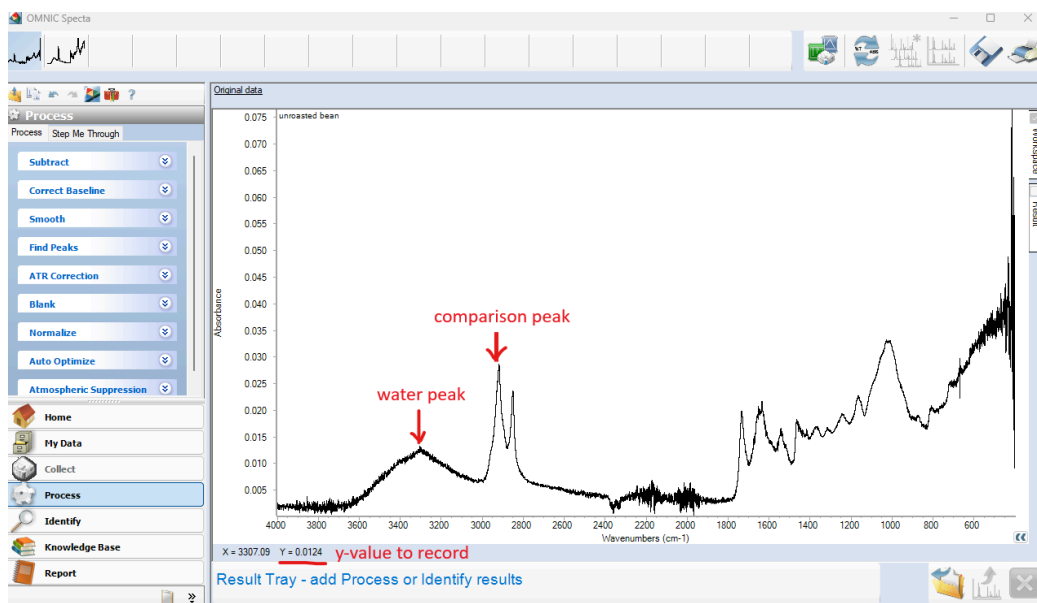


24. Click Process.

25. Hover your mouse over the peak of the water peak (a broad peak at about 3300 cm^{-1}). Record the y-value as the height of your peak.

26. Hover over the peak at 2900 cm^{-1} and record the y-value.

27. Divide the water peak height by comparing peak height to get a relative water peak height.



28. Repeat steps 23-27 with all of your spectra. Record the relative water peak heights for beans of different roast levels. Do you see a pattern? Is it what you expected?

Fourier-transform infrared spectroscopy (FTIR)

How does infrared spectroscopy work?

A beam of light in the infrared region of the electromagnetic spectrum is shined at a sample. Different types of bonds vibrate (by stretching, bending, wagging, etc.) at different wavelengths, which can be used to derive the types of bonds present in organic compounds.

How do you read an infrared spectroscopy graph?

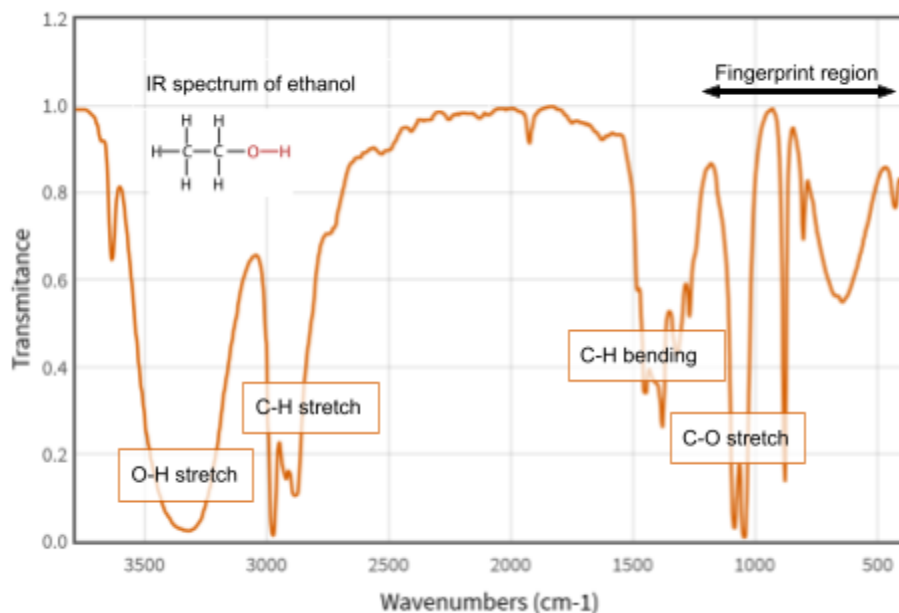


Table 1

Infrared absorption data

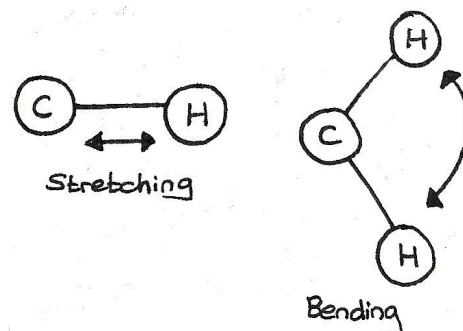
Bond	Wavenumber /cm ⁻¹
N—H (amines)	3300–3500
O—H (alcohols)	3230–3550
C—H	2850–3300
O—H (acids)	2500–3000
C≡N	2220–2260
C=O	1680–1750
C=C	1620–1680
C—O	1000–1300
C—C	750–1100

An IR spectroscopy graphs the sample's transmittance for each wavenumber. Wavenumbers are the reciprocal of the wavelength, i.e. $3000 \text{ cm}^{-1} = 3.33 \mu\text{m}$. Higher wavenumbers correspond to higher energy light being absorbed.

To determine a molecule's structure from its IR spectrum, determine the location of an absorption band (i.e. the dip at 3300 cm^{-1} above) and look at an infrared absorption data table. Looks like this corresponds to an O-H bond! Repeat for every notable dip on the spectrum and you'll find that the spectrum lines up with ethanol. There is a lot more nuance here, so I recommend reading up!

Where does the Fourier-transform part of the name come from?

When shining light at the sample, instead of iterating through each wavelength one by one (like UV-Vis spectrometers do), beams with different combinations of wavelengths are sequentially shined at the sample. Then, the computer takes the data and infers the absorptions at each wavelength using Fourier transform.



Instructions on using the Breakerspace FTIR Spectrometer

The Breakerspace has a Nicolet iS5 Fourier-transform infrared (FTIR) spectrometer, which is capable of analysing solids, liquids, gases, and thin films.



Instrument start-up

- Log on to instrument workstation using your MIT Kerberos
- Start OMNIC software
- Verify instrument connection via system status
- Remove protective cover from ATR crystal plate
- Clean ATR crystal
 - To clean the ATR crystal, wipe with isopropyl alcohol and a Kimwipe
- Collect background in the software

Operation

- Clean ATR crystal again if needed
- Place a sample over the ATR crystal
 - For *solid* samples
 - Ensure they are smooth and free of contaminants on the surface
 - For *powder* samples
 - Place enough powder to cover the ATR crystal completely.
 - Screw in either the concave or self-leveling pressure tip
 - Turn the black knob to push down on the sample until the clutch slips. The goal is to cover the surface of the ATR crystal completely!
 - For *liquid* samples:
 - Place a droplet on the ATR crystal.
 - Volatile samples should be covered with the volatiles cover.
- Collect sample in the software
- Save the sample spectrum, and repeat!

Instrument shut down:

- Save all data (each spectrum must be individually selected and saved)
- Close OMNIC and log-off the workstation
- Clean ATR crystal
- Put cover on crystal plate and clamp in place using pressure tower
- Do not switch the machine off (the manufacturer recommends leaving the instrument powered when not in use)

See the Breakerspace website for the extended SOP! Each workstation also has a desktop icon linking to its SOP.

As always, if you need help, contact a lab assistant!